

SUSHRUTH VELDI

Houston, Texas | sushruthveldi22@gmail.com | +1 7049231373 |

SUMMARY

Python Software Engineer with 2 years of experience building scalable backend systems, APIs, and cloud-native applications. Proficient in FastAPI, Flask, Django, and microservices architecture, with additional experience developing frontends using React.js. Skilled in ETL pipelines, database design, and DevOps (Docker, Kubernetes, AWS, GCP), delivering reliable, high-performance Python solutions in enterprise environments.

EXPERIENCE

TGS-NOPEC OIL & GAS

Houston, Texas

Data Science Software Developer

May 2025- Present

- Developed a React application with a FastAPI backend to validate SEGY and MDIO seismic data by generating trace maps, extracting spatial footprints, and performing geometric comparisons with reference data.
- Optimized performance using Ray for parallel processing (60% faster trace reading), containerized with Docker, and added interactive visualizations with GeoPandas, OpenCV, and Plotly.
- Automated validation workflows, leveraging LLMs for metadata extraction and structured storage, ultimately reducing 400+ man-hours annually and saving significant time and costs.
- Worked on extracting trace headers from seismic data, designing a schema using Pydantic, and storing the results in a structured format for easier access and retrieval.

The George Washington University

Washington, DC

Graduate Teaching Assistant – Linux for DevOps

Sep 2024-Jan 2025

- Led interactive lab sessions and hands-on exercises focused on Linux system administration, guiding students through essential tasks such as shell scripting, process management, user permissions, and networking configurations. Provided one-on-one support to help students grasp complex concepts and troubleshoot technical issues.
- Collaborated with the professor to design and grade assignments, ensuring fair and consistent assessment criteria. Provided constructive feedback to students, helping them improve their understanding of Linux and DevOps best practices.

Geo-Pragyan Tech

Hyderabad

Software Engineer

Aug 2021 – Sep 2022

- Developed and optimized Python automation pipelines and RESTful APIs for seismic data processing, streamlining workflows and reducing overall processing time by 35%.
- Designed and integrated React.js frontends with backend APIs, enhancing responsiveness of data-driven dashboards and improving user interaction speed while reducing latency by 25%.
- Built geospatial visualization tools using Pandas, NumPy, and OpenCV to analyze large datasets, cutting manual analysis efforts by 30% and enabling quicker decision-making.
- Containerized and deployed applications using Docker on AWS, improving scalability and system reliability while reducing production downtime by 20%.
- Collaborated with cross-functional teams in an Agile environment, contributing to sprint planning, code reviews, and deployments that improved release cycle efficiency by 15%.

Verzeo

Bengaluru

Software Engineering Intern

May 2021 - July 2021

- Assisted in developing Flask-based backend services for healthcare applications, contributing to coding, testing, and debugging tasks.
- Built data validation scripts to improve reliability and gained hands-on experience with Git and deployment workflows.

TECHNICAL SKILLS

- Programming & Scripting Languages:** Python, R, Java, SQL, HTML, CSS, JavaScript, Bash, Kotlin
- Data & Machine Learning Tools:** Tableau, Power BI, Excel, NumPy, Pandas, Scikit-learn, TensorFlow, OpenCV, MongoDB, MySQL, SQLAlchemy, Apache Spark, Hadoop, Datawrapper
- Frameworks & Cloud:** Fastapi, React, AWS, GCP, Azure Devops
- Operating Systems & DevOps:** Windows, Ubuntu, Git, Kubernetes, Docker

EDUCATION

The George Washington University

Washington, DC

Master of Science in Data Science

January 2025

Jawaharlal Nehru Technological University

Hyderabad, India

Bachelor of Technology in Computer Science

August 2022

CERTIFICATIONS

AWS Certified Cloud Practitioner – Amazon Web Services (AWS) | Issued: Oct 2024